Agenthon 2026

T2 Quant V4

Text-Conditioned Tail Mixture + Stress-Gating Report

Core question	Can the strong PCA2 marginal distribution be retained while a text-conditioned stress mixture improves tail calibration without damaging joint dependence?
Empirical V4 meaning	V4 is the text-tail / stress-mixture branch built on the PCA2 numeric backbone. It is not the broad roadmap label used in the earliest planning notes.
Final V4 candidate	V4.3: PCA2 base + text-conditioned stress mixture with a conservative 5%-10% tail-probability range.

What this report should make clear

- Why V4.0 failed when the natural PCA2 tail was physically truncated.
- How V4.1 corrected the mixture by keeping the full base distribution alive.
- How the AI/text layer was restricted to a structured signal rather than direct price forecasting.
- Why historical text coverage became a practical validation constraint.
- What V4.2b and V4.3 showed in multi-seed cross-unit testing.
- Why V4.3 was frozen as a conservative overlay before moving to joint-dependence research.

Core V4 idea	Do not ask the text model to generate the forecast distribution. Let the numeric PCA2 engine generate the distribution and use text only to modulate the probability and direction of an explicit stress component.

All results in this report are local pseudo-out-of-sample research results. They are not sealed leaderboard scores.

1. Starting point – PCA2 became the numeric backbone

V3 showed that factor-space simulation was more stable than the earlier bootstrap and t-copula experiments, but the Reference Gaussian still retained an advantage in the joint component.

Model	Marginal	Joint	Tail	Composite
Reference Gaussian	1.1858	4.4381	0.1579	1.9559
PCA2 r10/m60	1.1409	4.6291	0.1291	1.9850
Quant block1	1.1477	4.6657	0.1268	1.9989

Interpretation PCA2 already had strong marginal and tail behavior. V4 therefore avoided rebuilding the numeric core and focused on whether an explicit stress mixture could improve rare-event behavior while preserving the ordinary PCA2 distribution.

Forecast distribution = PCA2 numeric backbone + explicit stress-mixture overlay

2. V4.0 – the first tail-gating design failed

The first V4 implementation separated a core pool from an extreme PCA-factor pool by removing the largest factor-radius observations from the ordinary distribution and then injecting them only through a stress mixture.

2-fold diagnostic result

Model	Composite
Reference Gaussian	2.7017
PCA2 base	2.8787
Quant block1	2.9320
V4 tail 15%	3.6777
V4 core only	3.7038

Failure mechanism The stress component was not the only problem. Removing natural extremes from the base PCA2 pool made the ordinary distribution too thin. Tail calibration deteriorated even before the stress gate could help.

The V4.0 lesson was structural: a scenario overlay should add conditional stress mass, not delete the empirical tail that already exists in the base model.

3. V4.1 – preserve the base tail and add stress softly

V4.1 corrected the architecture. The base PCA2 distribution was left untouched, and the stress component became an additive mixture rather than a replacement for natural extremes.

$$F_t = (1 - p_{tail})\, F_{base} + p_{tail}\, F_{stress}$$

F_base	Ordinary PCA2 r10/m60 predictive distribution; no truncation.
F_stress	An ordinary PCA2 path plus one historical extreme PCA-factor event, spread over four days with decaying weights.
p_tail	Stress-mixture probability. Initially tested as fixed 0%, 2%, 5%, 10%, and 15%.
Direction	upper, lower, or both in target units.

Model	Marginal	Joint	Tail	Composite
Reference Gaussian	1.1858	4.4381	0.1579	1.9559
V4.1 fixed 10%	1.1397	4.6178	0.1272	1.9806
V4.1 fixed 5%	1.1391	4.6198	0.1295	1.9814
V4.1 fixed 15%	1.1417	4.6211	0.1296	1.9831
PCA2 base	1.1409	4.6291	0.1291	1.9850

Result V4.1 worked as an overlay: fixed 10% improved PCA2 slightly on marginal, joint, tail, and composite. The gain was modest, but the mechanism was now directionally correct.

4. V4.2 – text becomes a structured tail signal

The next step was to make the stress probability conditional on text without allowing the language model to choose prices, covariance matrices, or Monte Carlo paths.

AI output field	Meaning
shock_direction	upper / lower / both / none, expressed in target units
shock_severity	0 to 1 estimate of tail-event magnitude
confidence	0 to 1 confidence in the textual inference
immediacy	0 to 1 relevance to the forecast horizon
regime	descriptive state label used for diagnostics

$$\text{activation} = \text{severity} \times \text{confidence} \times \text{immediacy}$$

$$p_{\text{tail}} = \text{dormant} + (\text{maximum} - \text{dormant}) \times \text{activation}$$

Leakage rule At a historical walk-forward origin t, the prompt may contain only documents timestamped at or before t. Missing historical text is treated as missing evidence; it is never fabricated.

This provider separation also matches the competition environment: the numeric forecast engine is deterministic given the structured signal, while the approved House Model or another adapter can supply that signal.

5. Historical text coverage became the binding constraint

The original F3 election unit had no eligible historical text for its 24 walk-forward origins, so V4.2 could not be honestly evaluated there. A coverage scan was used instead of fabricating signals.

Rates-related unit	Text-covered folds
t2-F4-cpi-friday-2022	5 / 24
t2-F4-funding-stress-10y-2008	2 / 24
t2-F4-ust10y-supply-2023	2 / 24
t2-F1-liftoff-telegraph-2015	1 / 24
t2-F2-taper-testimony-2013	1 / 24
t2-F4-covid-rates-2020	1 / 24
t2-F4-svb-whiplash-2023	1 / 24
t2-F4-ust-downgrade-watch-2011	1 / 24

Methodological decision

Validation was restricted to folds where the historical prompt actually contained eligible text. The CPI-Friday unit was used first because it had the highest coverage, but its five folds were treated as a small diagnostic rather than a definitive benchmark.

For CPI-Friday, only the first two origins received an active upper-tail signal; later origins used the dormant floor because the available FOMC document had become stale relative to the forecast origin.

6. V4.2b – raising the dormant floor from 2% to 5%

The first 2%-15% text gate was almost tied with fixed 10%. A single coarse change was tested: preserve the same text signals, but raise the dormant floor to 5%.

Model	Composite on 5 CPI folds
V4.2b gate 5%–15%	0.246285
V4.1 fixed 10%	0.246978
V4.2 gate 2%–15%	0.247039
PCA2 base	0.248410
Reference Gaussian	0.250421

Small-sample result The 5%-15% mapping was best on these five folds, but the sample was too small for fine tuning. The mapping was frozen and moved to cross-unit, multi-seed validation.

This was an important discipline choice: no 5%-9%, 5%-11%, 5%-12% grid was searched after observing the five-fold result.

7. V4.2b cross-unit multi-seed test

The same filled text signals were evaluated on 15 text-covered folds across nine rates-related units using 5,000 draws and five RNG seeds.

Mean fold-weighted delta vs fixed 10%	-0.4178% (text gate better)
Mean unit-equal delta vs fixed 10%	-0.0457% (essentially tied)
Seeds with fold-weighted improvement	5 / 5
Seeds with unit-equal improvement	3 / 5

Unit	Mean % delta vs fixed10
CPI Friday 2022	-1.2455%
UST10Y Supply 2023	-1.0342%
Taper Testimony 2013	-1.0159%
COVID Rates 2020	-0.0587%
SVB Whiplash 2023	+0.1085%
Funding Stress 2008	+0.1605%
Liftoff 2015	+0.1642%
Considerable Removal 2004	+0.2517%
UST Downgrade Watch 2011	+2.2577%

New finding Text contained useful information on average, but mapping textual severity linearly into stress probability was too aggressive in some episodes. The UST downgrade-watch fold was the clearest failure.

8. V4.3 – conservative 5%–10% activation cap

V4.3 kept every text signal fixed and changed only the maximum stress probability from 15% to 10%. This was a coarse shrinkage step, not a new parameter search.

V4.3: p_tail in [5%, 10%]

Metric	V4.3 result
Mean fold-weighted % vs fixed10	-0.3995%
Mean unit-equal % vs fixed10	-0.1284%
Mean fold-weighted % vs V4.2b	+0.0237%
Mean unit-equal % vs V4.2b	-0.0772%
Seeds beating fixed10, fold-weighted	5 / 5
Seeds beating fixed10, unit-equal	4 / 5

Interpretation V4.3 did not clearly dominate V4.2b, but it reduced the downside from over-activating stress. The downgrade-watch loss fell from about +2.26% under 5%-15% to about +0.95% under 5%-10%.

9. V4 final decision and handoff to V5

V4 established a workable text-tail overlay, but it did not eliminate the larger dependence problem. The final V4 decision was therefore conservative.

Question	V4 answer
Keep the PCA2 base tail?	Yes. Never truncate the natural base tail.
Let AI generate price paths?	No. AI returns only structured stress information.
Use a dormant stress floor?	Yes. 5% was more robust than 2% in the text-covered sample.
Maximum text activation?	Freeze at 10% for the conservative V4.3 candidate.
Did V4 solve Joint?	No. Joint dependence remained the larger modelling bottleneck.
Next step	V5: isolate and diagnose the joint copula/dependence layer.

One-line conclusion

V4 succeeded as a controlled text-conditioned tail overlay: preserve the PCA2 base distribution, add stress softly, and shrink text activation. The remaining research problem moved back to joint dependence.